

Overviews on adverse events in pharmacological interventions

Ms. Elena Dorando

Institute for Research in Operative Medicine, Witten/Herdecke University, Germany
Elena.Dorando@uni-wh.de

Mr. Thilo Sachse

Institute for Research in Operative Medicine, Witten/Herdecke University, Germany
Thilo.Sachse@hhu.de

Dr. Tim Mathes

Institute for Research in Operative Medicine, Witten/Herdecke University, Germany
Tim.Mathes@uni-wh.de

Ms. Simone Heß

Institute for Research in Operative Medicine, Witten/Herdecke University, Germany
Simone.Hess@uni-wh.de

Prof. Dr. Petra Thürmann

Philipp Klee-Institute of Clinical Pharmacology, Helios University Hospital Wuppertal,
University Witten/Herdecke, Germany
petra.thuermann@uni-wh.de

PD Dr. Sven Schmiedl

Philipp Klee-Institute of Clinical Pharmacology, Helios University Hospital Wuppertal,
University Witten/Herdecke, Germany
sven.schmiedl@uni-wh.de

Dr. Salmaan Kanji

The Ottawa Hospital and Ottawa Hospital Research Institute, Ottawa, Ontario, Canada
skanji@toh.ca

Dr. Carole Lunny

Cochrane Hypertension Group and the Therapeutics Initiative
Department of Anesthesiology, Pharmacology and Therapeutics, University of British
Columbia, Vancouver, Canada
Carole.lunny@ti.ubc.ca

Mr. Pierre Thabet

Hôpital Montfort, Ottawa, Ontario, Canada
pierrethabet@montfort.on.ca

Dr. Dawid Pieper

Institute for Research in Operative Medicine, Witten/Herdecke University, Germany
Dawid.Pieper@uni-wh.de

Introduction

Background

Overviews of reviews systematically compile evidence of multiple systematic reviews in order to answer questions regarding efficacy, effectiveness and safety of health care interventions ¹. Sometimes they are also called “umbrella reviews”, “reviews of (systematic) reviews” or “meta-reviews”. Following Cochrane guidance, we henceforth refer to them as “overviews”.

Overviews can provide support for various decision makers, including clinicians, policy makers and the patient ¹. Furthermore, the data-aggregating approach of overviews can be helpful to identify rare events that otherwise would not be detected in a single systematic review or a single study.

According to Cochrane, conducting an overview of multiple systematic reviews is appropriate for four reasons: First, to summarize the evidence of different interventions for the same condition, second, to summarize the evidence for different outcomes of the same intervention and condition, or third, to summarize the evidence of the same intervention for different conditions or populations. The fourth reason for conducting an overview is to summarize evidence regarding the prevalence of adverse effects or adverse events from multiple systematic reviews ¹. We will focus our research on such overviews on adverse events and effects in a pharmacological context.

An adverse event (AE), or “adverse experience”, is defined by the WHO as “any untoward medical occurrence that may present during treatment with a medicine, but which does not necessarily have a causal relationship with this treatment” ². However, there are several other terms that are also used in the context of safety and potential harms of healthcare interventions. An adverse effect, for example, is defined as “a response to a medicine which is noxious and unintended, and which occurs at doses normally used in man” according to the WHO³. Interestingly, the term “occurring at doses normally used in man” is not employed by the European Medicines Agency. Other synonyms used are “adverse reaction” or “adverse drug reaction”. According to the Guideline on good pharmacovigilance practices (GVP), an adverse (drug) reaction is “a response to a medicinal product which is noxious and unintended [DIR 2001/83/EC Art 1(11) ⁴]. Response in this context means that a causal relationship between a medicinal product and an adverse event is at least a reasonable possibility (see GVP Annex IV, ICH-E2A Guideline ⁵)” ⁶. An adverse reaction, in contrast to an adverse event, is characterized by the fact that a causal relationship between a medicinal product and an occurrence is suspected. Furthermore, the term “side effect” denotes any (adverse or beneficial) unintended effect of the drug and can be dose related or not. A toxic effect, on the other hand, is always dose related ⁷. It should be noted, that between 20 % and 50 % of adverse (drug) reactions are preventable, i.e. there is an underlying medication error (e.g. patient took the wrong dose, physician prescribed a too high dose). However, a drug frequently associated with medication errors is apparently not a “safe drug”. The term “complication” is usually used for adverse events or effects caused by surgical or other invasive interventions ¹. In the interests of simplification, we will refer to all those terms as adverse events in the following. The non-standardized use of terms is known to be problematic in this field anyway.

While methods for conducting overviews in general have evolved in the recent years ⁸⁻¹², overviews on adverse events have not yet been analyzed in detail.

Objectives

There are two main objectives. That is 1) to investigate reporting and methodological characteristics of overviews on adverse events, and 2) to summarize the evidence on adverse events from published overviews. While the first objective will aim to improve the methods of such overview in a long-term, the second objective will resemble a scoping review.

Methods

Eligibility criteria

Study designs

We will include overviews that have systematically searched at least one database for systematic reviews. If updates of overviews have been published, we will use the most recent version. We will exclude protocols of overviews, and overviews which are used to inform clinical practice guidelines or health technology assessments.

Language

We will include overviews written in English, French or German.

Publication date

There will be no restrictions due to date of publication.

Publication status

Only published overviews will be included.

Participants / Patients

There are no restrictions with respect to characteristics of participants, while only human studies will be included.

Interventions being studied

Overviews on pharmacological interventions (administration of one or more drugs) will be included.

Outcomes

It can be assumed that the vast majority of overviews (irrespective of the reason they are conducted for) will report any adverse events if reported in the included systematic reviews. Here, we are specifically interested in overviews specifically conducted for the fourth reason (see above), that is to summarize evidence regarding the prevalence of adverse effects or adverse events from multiple systematic reviews. Therefore, we will only include overviews designed specifically to investigate adverse events.

Information sources and search strategy

We will search Medline, EMBASE, and the CDSR. For our search in Medline via OVID we will use a validated search filter for overviews¹³ in combination with a search filter for adverse events¹⁴. An outline of the search strategy developed by an information specialist (SH) can be found below. As the search strategy is based on the combination of two validated search filters, we do not feel it is necessary that the search strategy is checked by another information specialist using the PRESS guideline¹⁵. The search strategy will be adapted to EMBASE (via Elsevier). The CDSR via Cochrane Library will be searched using the term "overview" in title, abstract, and keywords. In addition, we will search Epistemonikos. There will be no restrictions due to publication period, but only published overviews composed in English, French or German will be included. We will not search for grey literature.

Search strategy Medline via Ovid:

((overview\$ or review or synthesis or summary or Cochrane or analysis) and (reviews or meta-analyses or articles or umbrella)).ti. or "umbrella review".ab. or (meta-review or metareview).ti,ab. and ((ae OR co OR de).fs. OR (safe OR safety OR side effect* OR undesirable effect* OR treatment emergent OR tolerability OR toxicity OR adrs OR (adverse adj2 (effect OR effects OR reaction OR reactions OR event OR events OR outcome OR outcomes))).ti,ab.)

Data management

Microsoft Excel will be used to manage and analyze the results of our search.

Selection process

Titles and abstracts retrieved from the search will be screened for eligibility by one review author and those that are not considered relevant will be evaluated by a second reviewer (liberal acceleration). Afterwards, full text screening will be independently conducted by two reviewers. Discrepancies will be resolved through discussion or by involving a third person.

Data collection process

Before the actual data extraction, we will test a data extraction form using a set of 5 overviews. This form will then be modified according to the test feedback. One reviewer will extract relevant data from all included overviews and a second reviewer will verify the results. Again, discrepancies will be resolved by consensus or by consulting a third person.

Data items

To facilitate the comparison of the methodological characteristics of overviews of adverse events with the results of previous research on overviews in general, we will mainly follow the work of Pieper et al.¹⁶ when selecting the methodological data to be collected.

Therefore we will collect the following data from the included overviews: Journal name, journal impact factor in the year of publication, year of publication, number of authors, country of corresponding author, registration (e.g., PROSPERO), protocol availability (as mentioned or referenced by the authors), inclusion criteria, primary outcomes, number of databases searched, search restrictions (language, date, publication status), selection methods, number of included studies (SRs and primary studies), type of primary studies

included in SRs, data extraction methods, quality and risk of bias assessments of individual studies (if presented/reported in the overview), quality and risk of bias assessments of systematic reviews, strategies to deal with discordant reviews, use of terminology, update strategies, strategies for dealing with overlapping reviews, grading of evidence, synthesis methods, analysis (qualitative vs. quantitative), presentation of results (tabular, graphical, verbatim), causality assessments/considerations, publication bias, source(s) of funding, and conflicts/declaration of interest.

In addition, we will collect the following content-related data: population, disease, (group) of drugs, route of administration, and all effect estimates of adverse events. These data will in particular inform the second objective.

Quality assessment

There is currently no tool to assess the methodological quality/risk of bias of overviews. However, we will use the PRIO (Preferred reporting items for overviews of systematic reviews) checklist ⁹ that was designed with a specific emphasis on harms in order to assess the reporting quality of the included overviews.

Data synthesis

For methodological and reporting characteristics, data will be analyzed descriptively. Data will be summarized as medians and interquartile ranges for continuous outcomes and as frequencies for categorical items.

We will present data as extracted from the overviews without re-analyzing data from overviews, SRs or trials. The presentation of outcome data will be decided depending on the available data. We expect the presentation of outcome data to be very heterogeneous. However, preference will be given to tables over a narrative description. Where appropriate, we will consider converting effect estimates to facilitate the interpretation of the data.

Further, we will present a map of the results to meet our second objective. Again, preference will be given to tables over a narrative description. Furthermore, we will divide our results into the following categories: population, intervention, and (type of) adverse event. We will leave the possibility open for additional analyses depending on our included sample of overviews.

We will create a citation matrix to assess whether overlapping SRs exist (i.e. SRs that are included in more than one overview). To guide the interpretation, we will calculate the corrected covered area (CCA) ¹⁷. In contrast to its original intent, the CCA will not be calculated at the overview level.

Discussion

Our review of overviews will provide a summary of all available evidence from overviews on adverse events in pharmacological interventions. This data should serve as an easy-front end for decision makers. Our review will also be able to guide the methodological development of overviews on adverse events that have been a rather neglected area so far.

We will mainly use passive dissemination strategies. In particular, we aim to publish our findings in a methodological and a pharmacological journal, respectively. We anticipate

using the results of this review to inform the development of guidelines to conduct and report these types of overviews.

References

1. Higgins JPT, Green S. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0. Updated March 2011. Accessed February 12, 2020.
2. WHO. *The importance of pharmacovigilance: Safety monitoring of medicinal products*. Geneva: World Health Organization; 2002.
3. WHO. *WHO Meeting on International Drug Monitoring: the Role of National Centres (1971: Geneva, Switzerland) & World Health Organization. (1971: Geneva, Switzerland)*: World Health Organization; 1972.
4. European Commission. Directive 2001/83/EC of the European parliament and of the council of 6 November 2001 on the community code relating to medicinal products for human use. <https://eur-lex.europa.eu/eli/dir/2001/83/2019-07-26>. Updated July 26, 2019. Accessed April 3, 2020.
5. European Medicines Agency. ICH Topic E 2 A: Clinical Safety Data Management: Definitions and Standards for Expedited Reporting. <https://www.ema.europa.eu/en/ich-e2a-clinical-safety-data-management-definitions-standards-expedited-reporting>. Updated 2006. Accessed April 3, 2020.
6. European Medicines Agency. Guideline on good pharmacovigilance practices (GVP) - Annex I - Definitions (Rev 4). <https://www.ema.europa.eu/en/human-regulatory/post-authorisation/pharmacovigilance/good-pharmacovigilance-practices>. Updated October 12, 2017. Accessed April 3, 2020.
7. Edwards IR, Aronson JK. Adverse drug reactions: definitions, diagnosis, and management. *The Lancet*. 2000;356(9237):1255-1259.
8. Ballard M, Montgomery P. Risk of bias in overviews of reviews: a scoping review of methodological guidance and four-item checklist. *Res Synth Methods*. 2017;8(1):92-108.
9. Bougioukas KI, Liakos A, Tsapas A, Ntzani E, Haidich A-B. Preferred reporting items for overviews of systematic reviews including harms checklist: a pilot tool to be used for balanced reporting of benefits and harms. *J Clin Epidemiol*. 2018;93:9-24.
10. Lunny C, Brennan SE, McDonald S, McKenzie JE. Toward a comprehensive evidence map of overview of systematic review methods: paper 1-purpose, eligibility, search and data extraction. *Systematic reviews*. 2017;6(1):231.
11. Lunny C, Brennan SE, McDonald S, McKenzie JE. Toward a comprehensive evidence map of overview of systematic review methods: paper 2-risk of bias assessment; synthesis, presentation and summary of the findings; and assessment of the certainty of the evidence. *Systematic reviews*. 2018;7(1):159.

12. Pollock M, Fernandes RM, Pieper D, et al. Preferred Reporting Items for Overviews of Reviews (PRIOR): a protocol for development of a reporting guideline for overviews of reviews of healthcare interventions. *Systematic reviews*. 2019;8(1):335.
13. Lunny C, McKenzie JE, McDonald S. Retrieval of overviews of systematic reviews in MEDLINE was improved by the development of an objectively derived and validated search strategy. *J Clin Epidemiol*. 2016;74:107-118.
14. Golder S, McIntosh HM, Duffy S, Glanville J. Developing efficient search strategies to identify reports of adverse effects in MEDLINE and EMBASE. *Health Info Libr J*. 2006;23(1):3-12.
15. McGowan J, Sampson M, Salzwedel DM, Cogo E, Foerster V, Lefebvre C. PRESS Peer Review of Electronic Search Strategies: 2015 Guideline Statement. *J Clin Epidemiol*. 2016;75:40-46.
16. Pieper D, Pollock M, Fernandes RM, Büchter RB, Hartling L. Epidemiology and reporting characteristics of overviews of reviews of healthcare interventions published 2012-2016: protocol for a systematic review. *Systematic reviews*. 2017;6(1):73.
17. Pieper D, Antoine S-L, Mathes T, Neugebauer EAM, Eikermann M. Systematic review finds overlapping reviews were not mentioned in every other overview. *J Clin Epidemiol*. 2014;67(4):368-375.